

Finding the Hypotenuse

Pythagoras' Theorem. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner builds a reliable, step-by-step method for finding the hypotenuse (the longest side, opposite the right angle) of a right-angled triangle when the two shorter sides are known. The emphasis is on **reasoning that the learner can reproduce on paper**, not on operating a machine that produces answers. Every step is one true line of working.

Driving Question: *How can I recognise the hypotenuse and build a correct step-by-step solution for its length?*

The mathematics, in plain terms

Pythagoras' Theorem says that in a right-angled triangle, the square on the hypotenuse equals the sum of the squares on the other two sides: $a^2 + b^2 = c^2$, where c is the hypotenuse. "The square on a side" means the area of a square whose side is that length. The method has six steps: (1) identify the hypotenuse; (2) pick out the two shorter side lengths; (3) write the theorem for this triangle; (4) square each shorter side; (5) add the two squared results; (6) take the square root of the total to get the length. Squaring turns lengths into areas; the final square root turns the area back into a length — which is why it comes last.

Why the steps are in that order

It can help to know *why* the method runs squaring, then adding, then rooting. A length on its own cannot simply be added to another to give the hypotenuse — the relationship lives in **areas**. Squaring each shorter side turns it into the area of a square. The theorem tells us those two areas add up to the area of the square on the hypotenuse. So we add the two areas. That total is an *area*, not yet a length — so the final square root converts it back into the length of the hypotenuse. If a learner stops at the addition, they have found an area and still need the root; if they add before squaring, they have skipped the move into areas altogether. Holding on to "lengths become areas, then back to a length" explains almost every slip in this topic.

The shape of the 45 minutes

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected and fine. The Cornerstones below are each given an honest mathematical role in this lesson:

Cornerstone	Its role here	Invitational timing
Connection	The hypotenuse is <i>connected</i> to the right angle — it lies opposite it; the theorem connects the three sides.	~8 min

Movement	Building the solution as a moving chain of true lines, one following from the last.	~22 min
Reflection	Asking of each line: is this true, and how do I know? Noticing the square-root step.	~4 min
Creativity	Making a teaching card or a worked example in the learner's own style.	~3 min
Rest	The deliberate pause before the square root; not rushing the final line.	~3 min
Nutrition	Intellectual (Mode B): this lesson feeds precision and the habit of checking each step is true.	~5 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is to ask a question that returns the thinking to the learner — never to supply the answer.

Naming the wrong side as the hypotenuse

The learner labels a side without checking the right angle. Repair prompt: “Which angle is the right angle? The hypotenuse is the side directly opposite it — and it is always the longest side.”

Adding the side lengths before squaring

The learner computes $a + b$ instead of $a^2 + b^2$. Repair prompt: “We add the *areas* of the squares on the sides, not the side lengths. What is the area of the square on this side?”

Forgetting the final square root

The learner offers c^2 as the length. Repair prompt: “You have found the area of the square on the hypotenuse. What length gives a square of that area?”

Squaring the sum instead of summing the squares

The learner computes $(a + b)^2$. Repair prompt: “Square each side on its own first, then add. What do you get for each square separately?”

Assuming a shorter side could be the hypotenuse

Repair prompt: “Can the longest side ever be one of the two that meet at the right angle? Where must the longest side be?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: **square numbers**, **square roots**, and **identifying the hypotenuse**. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Here is the kind of conversation that keeps the thinking with the learner. Imagine a triangle with a right angle, and the two shorter sides given. Notice that none of these prompts hands over a number — each one returns a decision to the learner:

- “Point to the right angle. Now run your finger to the side that does not touch it — what is that side called?”
- “We are going to work with the squares on the two shorter sides. What is the area of the square on this one?” (let them do each square separately)
- “What does the theorem tell us to do with those two areas?”
- “That total — is it a length yet, or an area? What one more move turns it into the length?”
- “How could you check whether your answer is sensible — should the hypotenuse be longer or shorter than each of the other two sides?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“What if their answer isn’t a whole number?”

That is normal and correct — most real triangles give a hypotenuse that is a rounded decimal. A sensible rounded length, with the working shown, is a strong answer.

“I can’t remember this maths myself — can I still help?”

Yes. Your most useful moves are asking the repair questions above, keeping the pace calm, and pointing to the built-in support. You do not need to know the answer to help a learner reason toward it.

“They got it wrong — should I correct it straight away?”

A wrong attempt is useful information and also counts toward unlocking the worked solution. Ask which step they are least sure of, rather than replacing their work.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These are descriptions of where a learner is right now, not labels of what they are. A learner who identifies the hypotenuse and sets up the theorem correctly is

already demonstrating real understanding, even if the arithmetic is still settling.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to a single, concrete triangle and the one question “which side is opposite the right angle?”. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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